



UNIVERSIDAD TECNICA FEDERICO SANTA MARIA

Tesis de Magister

Design and Sizing of an Energy Storage System for a Hybrid Tugboat

Tesis para optar al título de
Magister en Ciencias de la Ingeniería Electrónico

Alumno
Leonardo Solis Zamora

Profesor Guía
Dr. Marcelo Pérez Leiva

Comisión evaluadora
Nombre del primer correferente, Correferente, UTFSM
Nombre del segundo correferente, Correferente, CODELCO

Enero 2025 , Valparaíso, Chile

This is the dedicatory page.

Agradecimientos

This is the abstract

Resumen

This is the abstract

Abstract

This is the abstract

Índice general

Agradecimientos	ii
Resumen	iii
Abstract	iv
1. Introduction	1
1.1. Motivation and Background	1
1.2. Challenges and Research Opportunities	5
1.3. Thesis Objectives and Outline	5
Bibliografía	6

Índice de figuras

1.1. Global sources of greenhouse gas emissions. Source: theroundup.org	1
1.2. Breakdown of CO ₂ emissions from the energy sector and the transport subsector. Source: theroundup.org	2
1.3. The MARPOL Annex VI emission limits for global and Emission Controlled Areas (ECAs perspective (a) SO _x and (b) NO _x . (Source: DieselNet	3
1.4. IMO greenhouse gas emissions strategy. Source: Lloyd Register, NOV, Assessment of IMO Mandated Energy Efficiency Measures for International Shipping	4

Índice de tablas

Capítulo 1

Introduction

1.1. Motivation and Background

The growing global concern over climate change has significantly pressured industries to adopt sustainable practices. The maritime sector, responsible for approximately 2.5 % of energy sector CO₂ emissions, has continuously increased its emissions, as shown in Fig. 1.

<https://theroundup.org/co2-greenhouse-gas-emission-statistics/>

<https://www.container-xchange.com/blog/shipping-emissions/>

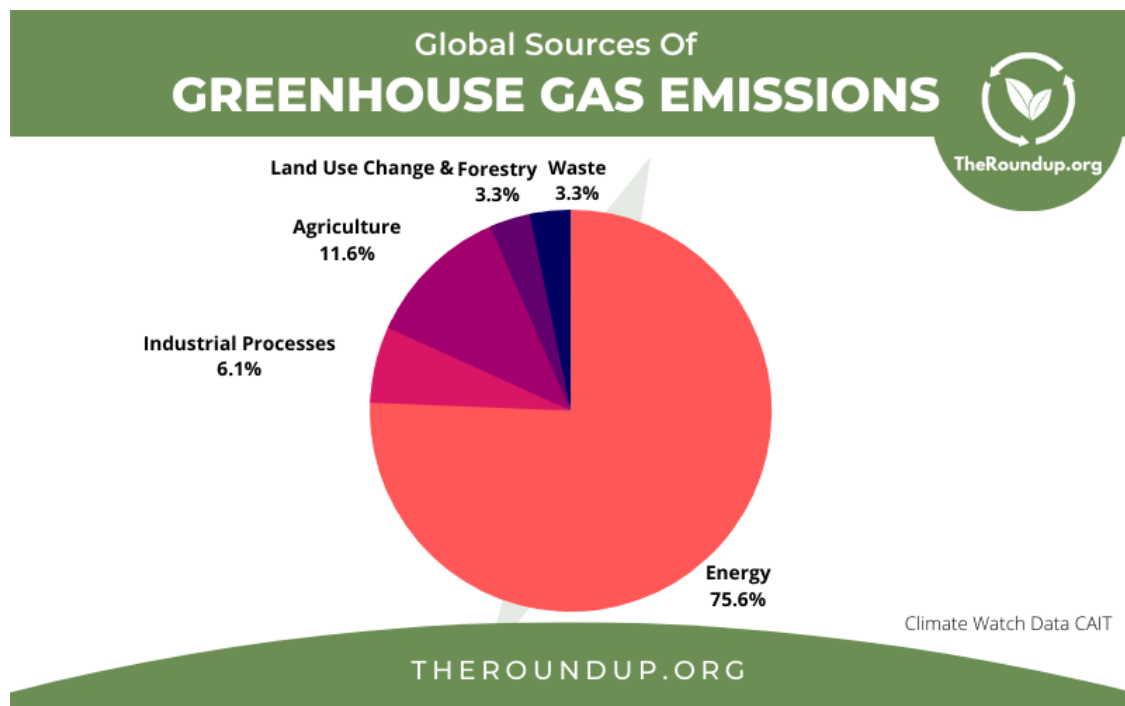


Figura 1.1: Global sources of greenhouse gas emissions. Source: theroundup.org

Addressing emission reduction is critical, as strict regulations on emissions and fuel efficiency aimed at mitigating the environmental impact of maritime activities are being implemented worldwide [2]. In the area of ships there are several activities to replace the proven diesel mechanical drives by new concepts. The triggers are firstly the ever increasing price of fossil fuels and the other a more stringent emission standard in so-called Sulphur Emission Control Areas

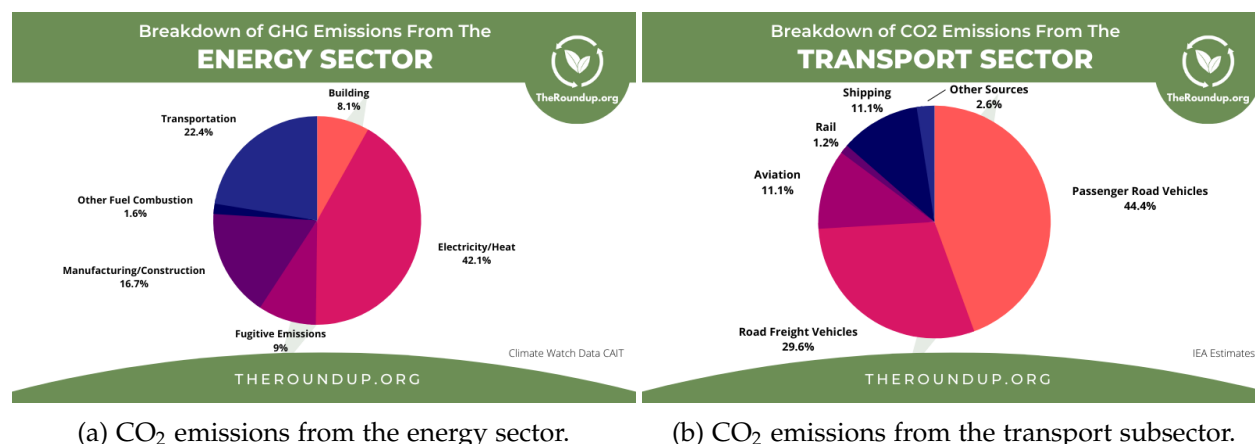


Figura 1.2: Breakdown of CO₂ emissions from the energy sector and the transport subsector. Source: theroundup.org

(SECA) from 1 January 2015. After that, only fuels may be used with a sulfur content of 0.1 percent in may coastal regions, which means additional costs for pushcase of fuel [2].

The International Maritime Organization (IMO) is implementing strategies to reduce the emissions from international shipping by establishing standards, imposing restrictions and providing incentives. The initial strategy envisages a reduction in total GHG emissions from international shipping by at least 50 % by 2050 compared to 2008. consistent with the Paris Agreement. IMO's current global standard for SO_x emissions is 0.5 % mass by mass (m/m) by 2025 per the International Convention for the Prevention of Pollution by Ships (MARPOL 73/78) Annex VI. Similarly, the emission limit for NO_x from engines is dependent on the type of the engine and the nominal engine speed. The global limit has been set to 14.4 g/kWh for engines with speeds less than 130 rev/min and 7.7 g/kWh for engines with speed greater than 2000 rev/min.

To comply with the standards and emissions reduction goals set by IMO, the maritime industry is looking for reduced and zero-emission technologies and approaches. The transition from traditional fossil fuel engines to zero-emissions technologies is challenging in a competitive market, where manufacturers are struggling to increase revenue and reduce production cost. The incentives to reduce emissions and improve the efficiency of equipment are motivating ship manufacturers to adopt zero-emission technologies. Fuel cells and energy storages (for example, batteries and supercapacitors) reduce GHG emissions and are the key enablers for the realization of low-emissions operation of marine vessels.

Electric and hybrid propulsion architectures, allowed by innovative and hig performance components (batteries, fuel cells, supercapacitors, etc) are a promising track to reduce strongly pollutants and GHG emissions [3]. New developments of ellectrically propelled ships are based on an integrated energy distribution system architecture that releases large propulsion machines [4].

Electric power is provided by generators operated by a gas turbine, a diesel engine, or a combination of both. The main advantages of full-electric propulsion include better dynamic performance (start-up,arrest, changes in speed, and regenerative braking), reduced fuel consumption due to the more efficient and flexible use of generators that work within their minimum specific fuel consumption region, and the possibility of having shorter shaft lines because only the electric motor must be installed close to the propeller.

The hybrid propulsion system uses a combination of diesel engines or gas turbines with electric motors to provide a wide range of propeller speeds. One of the main advantages of this

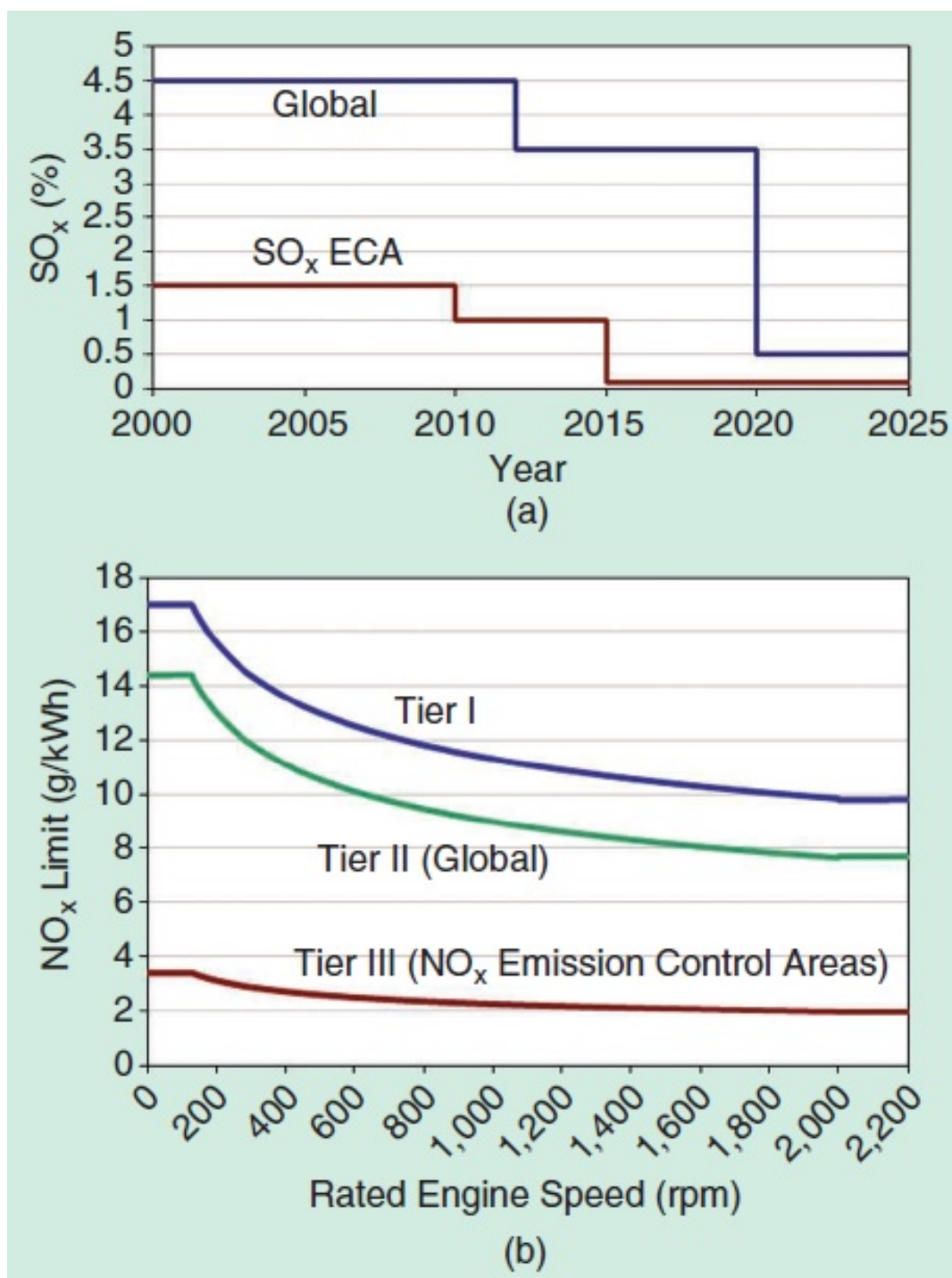


Figura 1.3: The MARPOL Annex VI emission limits for global and Emission Controlled Areas (ECAs perspective (a) SO_x and (b) NO_x . (Source: DieselNet

configuration concerns an increase in efficiency at low loads. Hence, less fuel consumption, fewer emissions, and a small footprint are achieved. Hybrid propulsion systems are interesting when there is a large variation in power demand during operations. The design rules of this type of systems can differ significantly for different types of vessel, mainly due to the different types of

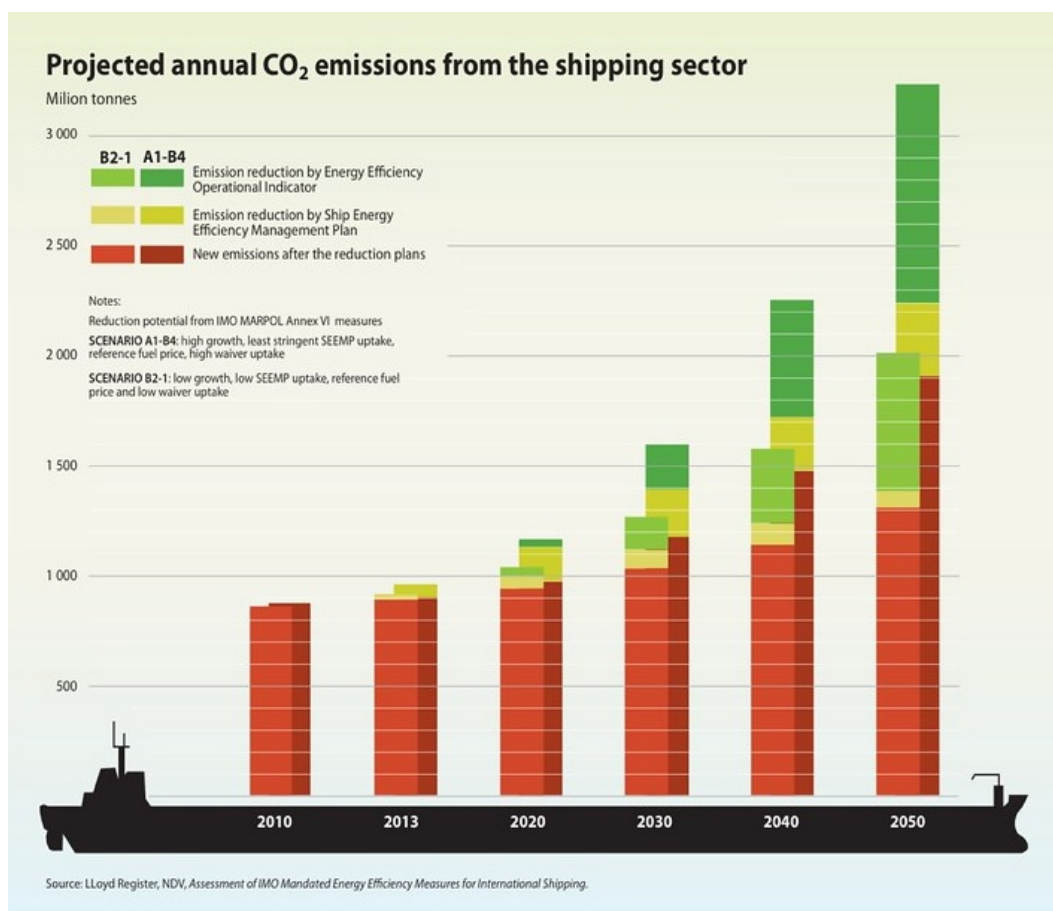


Figura 1.4: IMO greenhouse gas emissions strategy. Source: Lloyd Register, NOV, Assessment of IMO Mandated Energy Efficiency Measures for International Shipping

mission profiles and operation requirements.

Several studies have examined hybrid propulsion systems. A marine hybrid propulsion system, focusing on vector control of the electric motor during different modes and verifying the control feasibility [6]. The optimization of hybrid propulsion system design for a tugboat has been explored, presenting a methodology that streamlines powertrain component sizing and control, minimizing costs for a specific operating profile [7]. A coordinated control strategy for a variable-speed hybrid tugboat have been presented to improve fuel economy. The proposed strategy, validated through simulations and a smallscale experimental testbed, showed reduced costs and lower CO₂ emissions [8].

Among vessel types, tugboats—used for towing and maneuvering large ships—are among the highest emitters per unit of energy due to their highly variable load profiles [3]. Tugboats and ferries are ideal candidates for hybridization due to their operational profiles, which involve prolonged idling, low-speed maneuvering, and frequent speed changes that lead to inefficient fuel consumption [9].

This paper focuses on designing and sizing an energy storage system for a hybrid tugboat as a specific electro-mobility solution. Despite advancements in marine hybrid technologies, a standardized methodology for tugboat hybridization remains undefined [10]. The study presents a design methodology addressing parameters such as load profiles, power and energy demands,

battery technology selection, and propulsion system optimization. Theoretical analysis is presented and simulation shows a emissions reduction while maintaining the robustness needed for towing and transit activities.

1.2. Challenges and Research Opportunities

1.3. Thesis Objectives and Outline

[1],

Bibliografía

- [1] A. Carreno, M. Malinowski, M. A. Perez, and J. Ding, “Effects of grid voltage and load unbalances on the efficiency of a hybrid distribution transformer,” *IEEE Open Journal of the Industrial Electronics Society*, vol. 5, pp. 1206–1220, 2024.
- [2] Völker, Thorsten. “Hybrid Propulsion Concepts on Ships.” 2013.
- [3] N. Bennabi, J. F. Charpentier, H. Menana, J. Y. Billard and P. Genet, “Hybrid propulsion systems for small ships: Context and challenges,” *XXII International Conference on Electrical Machines (ICEM)*, Lausanne, Switzerland, 2016, pp. 2948-2954, doi: 10.1109/ICEL-MACH.2016.7732943.
- [4] C. A. Reusser, H. A. Young, J. R. Perez Osses, M. A. Perez and O. J. Simmonds, “Power Electronics and Drives: Applications to Modern Ship Propulsion Systems,” in *IEEE Industrial Electronics Magazine*, vol. 14, no. 4, pp. 106-122, Dec. 2020, doi: 10.1109/MIE.2020.3002493.